Huynh Le Dang Khoa

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Research Profile

Honors computer science graduate of Ho Chi Minh City University of Technology, Vietnam National University. My research direction is **computer vision and vision–language models**, and how that perception supports vision–language–action models for robot learning.
Recent experiments are in computer vision and robot simulation: reliable visual perception with foundation models, vision–language models, and early vision–language–action work, including learning from human video and evaluating robot behavior in simulation.

Education

Ho Chi Minh City University of Technology, Vietnam National University — Ho Chi Minh City
B.Eng. Computer Science & Engineering, Honors Program — *2022 – 2026*

- Excellent GPA in the Honors Program. Top 1% of the school. Official diploma expected November 2026.
- Undergraduate thesis on accuracy-aware scheduling for concurrent vision inference. Grade: **9.84/10**.
- Interests: computer vision; vision–language models; vision–language–action models; long-horizon robot learning; learning from human video.

Publications and Manuscripts

- *Reliable Few-Point Metric Grounding for Monocular Depth Foundation Models.* **ICLR 2027, under review.** On NYUv2 with UniDepthV2, eight sparse metric observations reduce AbsRel from 0.0766 to 0.0540 (**29.5%**); with 80% of the observations corrupted, AbsRel remains 0.0608.
- *Acquisition-Aware Anomaly Detection.* ***Image and Computer Vision*****, under review.** Introduces ARAD, a benchmark for anomaly detection under blur, motion, illumination, sensor noise, and mixed acquisition shifts, with a training-free detection framework.
- *Mutual Hand–Object Reasoning for Egocentric Manipulation Perception.* **ICRA 2027, under review.** Improves over CaRe-Ego by **6.93 mIoU** on the EgoHOS out-of-distribution split and by **3.82 mIoU** on mini-HOI4D without fine-tuning.
- *An Intelligent Virtual Machine Consolidation Approach for Energy-Efficient Cloud Data Centers.* **First author, ACOMPA 2025.** doi:10.1109/ACOMPA69200.2025.00021

Research Experience

ARENA Lab, Singapore Management University | SMART — Singapore / Remote
Research Collaborator, Eye Code Tech — *Feb 2026 – Present*
Advisors: Do Van Minh (SMART) and Prof. Linh Kästner (Doan Hoang Linh, ARENA Lab)

- Reliable visual reasoning with pretrained foundation models: few-point metric grounding for monocular depth, acquisition-robust anomaly detection, and egocentric hand–object perception.
- Current robot-learning directions: memory-augmented vision–language–action policies, demonstrations recovered from human video, and long-horizon manipulation.

HPCC Lab, HCMUT, Vietnam National University Ho Chi Minh City
Research Assistant *Jan 2025 – May 2026*
Advisors: Prof. Nam Thoai

- Thesis on scheduling concurrent heterogeneous vision inference under accuracy and service constraints.
- About **20% higher throughput** and **80% fewer SLA violations** than strong scheduling baselines on mixed workloads.
- Reproducible GPU benchmarks of latency, memory, and throughput for vision models.

Research in Progress

- **MemoryPi.** Explicit long-term memory for vision–language–action policies, so an action can depend on an observation or outcome no longer in the current frame. Motivated by long-horizon failures in RoboMME-style manipulation.
- **Video-to-simulation.** Recover object geometry, hand–object interaction, and demonstrations from monocular human video, then retarget the interaction into a robot simulator.
- **Generative evaluation.** Action-conditioned robot video and learned world models as an evaluation signal where authored 3D assets are insufficient. The open question is when a predicted visual future is reliable enough to judge a policy.

Additional Experience

Dien Toan Group Ho Chi Minh City
AI Engineer Intern *May 2025 – Aug 2025*

- Document-understanding and retrieval-augmented generation pipeline with multimodal language models, vector retrieval, PostgreSQL, MinIO, and Docker.
- Member of the team named Most Outstanding Intern Team.

Technical Preparation

- **Research stack.** Python, PyTorch, OpenCV, CUDA. Vision and multimodal foundation models, detection and segmentation, monocular depth, vision–language–action models, reinforcement learning.
- **Robotics.** Isaac Lab, manipulation evaluation, video-to-simulation, vision-based robot learning.
- **Also.** C/C++, Java, Go, SQL, Linux, Git.

Awards and Languages

- Most Outstanding Intern Team, Dien Toan Group. Second place, Innostart Entrepreneurship Competition. Top 4, UTE Junior Startup. Vietnamese (native). English (professional working proficiency).